Same job, higher wage? E-3 versus H-1B filings in US Labor Condition Applications, FY2017–2022

Group name	Ramanujan et al.
Dataset	kaggle.com/datasets/jishnukoliyadan/lca-programs-h1b-h1b1-e3-visa-petitions
Code	`https://github.com/sjwl07/h1b-e3-wage-gap`

#	Name	Roll no.	Institute email	Contribution
1	Shivam Jhunjhunuwala	2023505	shivam23505@iiitd.ac.in	Data loading, held-out split, report
2	Ramanujan Gopalan	2023424	ramanujan23424@iiitd.ac.in	Data dictionary, cleaning rules
3	Saanvi Singh	2023454	saanvi23454@iiitd.ac.in	EDA figures and captions
4	Mohammad Umar	2023324	umar23324@iiitd.ac.in	Research questions and test planning
5	Mayank Kumar	2023317	mayank23317@iiitd.ac.in	Hypothesis tests and checks

1. Dataset

Source: Kaggle, "LCA Programs (H-1B, H-1B1, E-3) Visa Petitions" (CC BY-NC-SA 4.0), taken from the US Department of Labor (DOL); six yearly CSV files, FY2017–2022, downloaded on 21 October 2026. Each row is one Labor Condition Application (LCA). Before hiring a foreign worker on an H-1B, H-1B1 or E-3 visa, an employer must file one, stating the job, where the person will work and the *prevailing wage*: the minimum pay DOL sets for that job, area and experience level. DOL then approves (certifies) or denies it.

Why it is raw: the decision labels are spelled differently before and after FY2020; FY2019's wage column holds the *offered* wage, not the prevailing wage; FY2019's work location is blank in 88% of rows; wages come in five pay units with typing errors (an "hourly" wage of 46,301, a yearly wage of $1 billion); 1,769 spellings of about 1,100 occupations; 255,125 spellings of employer names; 24% of rows are exact copies of another row.

	Raw files	After encoding
Rows	3,973,349	3,178,679
Columns	11–12	210
Rows × columns	45,736,562	667,522,590

Size check: the encoded count uses only the 80% exploration set (the other 20% is held back for testing, Section 2): $3{,}178{,}679 \times 210 = 667{,}522{,}590 \geq 3{,}000{,}000$. The 210 columns are 197 yes/no (one-hot) columns (the 50 most common occupations plus `Other`, work and employer state, year, quarter, visa, decision, pay unit, full-time, region) and 13 number columns. The raw files alone already pass: FY2017–19 have 11 columns and FY2020–22 have 12, so $1{,}943{,}626 \times 11 + 2{,}029{,}723 \times 12 = 45{,}736{,}562$.

Key features

(all columns, with type, unit, range and missing values by year: `report/data_dictionary.md`)

Feature	Type	Meaning
`log_pw`	continuous	log of the yearly prevailing wage ($15k–$462k after cleaning); outcome for P1
`case_status`	nominal	DOL's decision: certified (90.9%), certified then withdrawn, withdrawn, denied (0.9%); `denied` is the outcome for P2
`visa_class`	nominal	H-1B (97.6%), E-3 for Australians (2.0%), H-1B1 for Chile and Singapore (0.4%)
`soc_title`	nominal	occupation, 1,123 after cleaning; 30% are "Software Developers, Applications"
`worksite_state`	nominal	state where the person will work, 55 states and territories
`fiscal_year`	discrete	US fiscal year of the decision, 2017–2022
`employer_key`	nominal (ID)	cleaned employer name: 165,799 in all six years, 132,485 in the exploration set; used to group filings, not as a column
`employer_volume`	discrete	added by us: filings by that employer that year, 1–21,448
`full_time`	nominal	full-time job (98.2%)

2. Data preparation

Every rule was worked out on the exploration set and saved; the test set was later cleaned with the same saved rules, never re-worked. No rows are dropped: every row still has a valid decision, so a bad wage is set to missing (and marked) instead.

Split, before looking at the data: 80% for exploration, 20% held back for testing, seed 42. Ideas for tests come from looking at the data, and testing an idea on the data that gave it is circular, so all EDA and cleaning rules use only the 80%. All filings of one employer go to the same side, because they are near-copies and would otherwise appear in both; each side keeps the same mix of years and decisions (stratified group k-fold).

Problem	What we did	Why (and not the alternative)
decision spelled two ways (capitals before FY2020, normal case after)	mapped to 4 labels	otherwise one decision counts as two
FY2019 wage column is the offered wage	kept as its own column; FY2019's prevailing wage left blank	39% of FY2019 values are round thousands, against under 1% in every other year, and the median wage per occupation jumps $+15.6\%$ into FY2019 then falls -7.8% out of it (`report/fy2019_wage_check.md`)
777,688 exact copy rows (24%)	**kept**, and counted in `n_identical`	there is no case number, and in DOL's own data every row is a separate application. The copy share rises with employer size (18% $\rightarrow$ 55%) and differs by decision (36% certified vs 16% withdrawn); an export mistake wouldn't do that. Dropping them would delete real filings of big employers; their similarity is handled in the tests (Section 5)
five pay units, 427 wrong ("Hour" $= 46{,}301$)	427 changed to Year, then all turned into yearly pay ($\times$2080, 52, 26, 12)	a unit is wrong when value $\times$ unit is over $1M a year but the value alone is a normal salary ($15k–$1M): the unit is wrong, not the wage, so fixing it keeps the wage. One yearly scale makes all wages comparable (2,080 h = DOL's full-time year)
101 wages of $0 or with no unit	set to missing	impossible values; the row's decision is still valid
1,769 occupation spellings (codes stuck on, typos, plurals)	merged into 1,123: compare letters only, ignore plurals, then join rare spellings that are at least 92% similar	exact matching keeps e.g. "15-1121.00Computersystems-analysts" as a separate job. All 215 merges are listed in `report/soc_fuzzy_merges.md`; a looser cut-off would join "Electrical" and "Electronics" engineers (0.89 similar)
extreme wages (a $1B "yearly" wage)	IQR outlier rule (1.5) on *log* wage, separately for each occupation; 21,579 wages set to missing and marked	log, because wages have a long right tail; per occupation, because $200k is normal for a surgeon, not a cook. Blanking the wage (not the row) keeps the decision
employer country blank (100,932) or "Georgia"	set to USA when the employer's state is a US state; 16 stay blank	the address shows the country; a US state was typed as the country
FY2019 work location 88% blank	marked, not filled in	work and employer state match in only 52% of rows, and in FY2019 the blank rows are different (93% vs 60% certified), so filling them in would invent locations
columns to encode	year, quarter, decision, visa, pay unit, full-time, states, region as yes/no columns; flags 0/1; employer filing count logged	yes/no columns assume no order, so wages can move unevenly by year; the two outcomes stay as numbers (`log_pw`, `denied` 0/1) for the tests; filing counts are very uneven (1–21,448)
text columns with many values	yes/no columns only for 10 columns with few values; occupation as top 50 + `Other` (87.7% of rows); employer and job title not turned into columns	132k employers and 375k job titles would add half a million almost-empty columns; they enter as filing count and seniority flags instead

3. What changed after preparation

Exploration set, before and after cleaning (`report/cleaning_log.md`).

Measure	Before	After
Rows	3,178,679	3,178,679 (none dropped)
Columns	12	40 cleaned; 210 after encoding
Usable prevailing wages	2,645,476	2,623,796 (21,680 set missing: 21,579 outliers, 101 invalid)
Wage range (yearly)	$0 – $1B	$15,080 – $462,342
Pay units	5, 427 wrong	1 (yearly)
Occupation spellings	1,769	1,123
Job title spellings	439,495	375,194
Employer name spellings	204,209	132,485 employers
Blank employer country	100,932	16
Exact copy rows	777,688	777,688, kept and counted

Cleaning turned the wage into one comparable yearly number: the lowest cleaned wage is exactly $\$7.25 \times 2{,}080$, the US minimum wage for a full-time year, a sign the unit fixes worked. Its log is almost symmetric (skew -0.04, against 1.14 in dollars; Fig. 1). Merging occupation and employer spellings is what lets Section 5 compare filings within the same job and the same employer. **What we lost:** 21,680 wages (0.8%) set to missing; all of FY2019 (17% of rows) for wage questions, as it has no prevailing wage; the detail of the 1,073 rarer occupations, grouped as `Other` (12.3% of rows) in the encoding; and the difference between merged spellings, if a merge was wrong.

4. Basic questions and problem statement

Basic questions

	Question	Answer (exploration set)
1	Which employers file the most?	Infosys, TCS, Google, Amazon, EY; the top 10 file 16%
2	Which jobs?	software: computer jobs are 65%, one title alone 30%
3	How many filings are approved?	91% certified; only 0.87% denied
4	Have wages risen?	median prevailing wage $+33\%$, $72k (FY2017) $\rightarrow$ $96k (FY2022)
5	Do some regions pay more?	West $105k, South $83k, Midwest $78k

Our problem statements

P1 (primary). For the same occupation, state and year, do E-3 (Australian) filings have a different prevailing wage from H-1B filings, and does the difference remain within the same employer?

P2. Are E-3 filings denied more or less often than H-1B filings from employers of the same size in the same year, and within the same employer?

Why they are new: the basic questions look at one column at a time. DOL sets the prevailing wage from the job, area and experience level; the visa is not part of it, so "no difference" is the natural starting point for P1. A difference would mean employers file the two groups at different pay levels, which the data shows only indirectly (Fig. 4). P1 holds fixed what the basic questions show moves wages most (job, place, year), and then the employer. P2 asks whether the raw denial gap (E-3 denied 5.8$\times$ as often in the exploration set, 6.9$\times$ in the test set) survives employer size, the main other factor EDA found (Fig. 3). It is not a Simpson's paradox: E-3 is denied more in every size group too, so the question is how much of the gap size explains.

Correlation, not causation: both statements compare *filings*; they don't claim the visa causes anything. Australians may simply be hired into more senior roles, and worker experience and education aren't in the data. To claim cause we would need worker-level data to compare similar workers, or a rule change that affects only one visa, studied before and after. We have neither, so we make no causal claim.

5. Statistical inferences

Rows tested: the 20% held back: 794,670 filings from 33,314 employers never seen during exploration, cleaned with the saved rules; E-3 vs H-1B only (H-1B1 is 0.4%, a separate program). Wage tests use FY2017–18 and FY2020–22 (FY2019 has no prevailing wage) and rows with a valid wage and work state.

Common design, fixed before the test data was opened: each test is β in $y_i = \beta\,\mathrm{E3}_i + \alpha_{c(i)} + \varepsilon_i$, where α_c is a separate level (*fixed effect*) for each comparison cell, e.g. one occupation in one state in one year. Each choice has one reason:

- **One level per cell**: β is the E-3 minus H-1B gap *inside* cells, so job, place and year can't create it. It is a two-sample comparison done inside each cell and then averaged, like a paired t-test with groups instead of pairs.
- **Standard errors grouped by employer**: one employer's filings are near-copies (Section 2), not independent, and ignoring that makes the SE too small. Employers are the independent units, so the t-test uses employers $-\,1$ degrees of freedom.
- **Holm correction**: four tests at 5% each give up to a 19% chance of a false alarm; Holm keeps it at 5% overall.
- **Smallest meaningful difference (MID)**: with this many rows even a tiny gap is significant, so the 95% CI decides: *meaningful* (CI beyond the MID), *possibly too small* (CI includes it) or *negligible* (CI inside it).

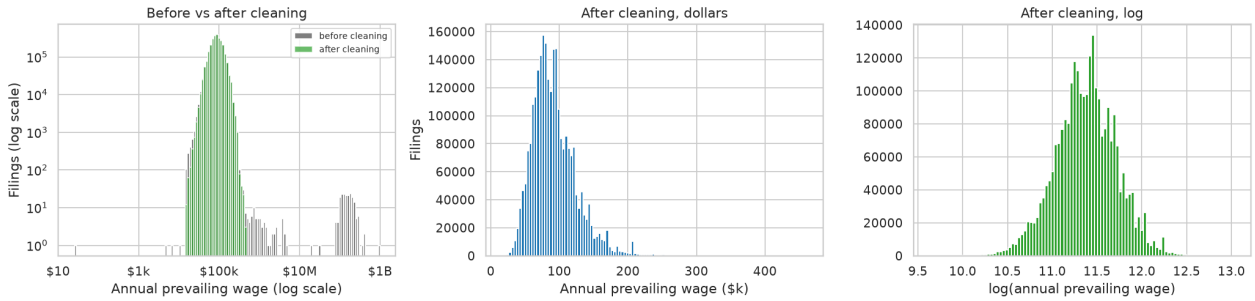


Figure 1: Cleaning removes impossible wages: before, $25 to $1B a year (grey); after, $15k–$462k, with a symmetric log (right).

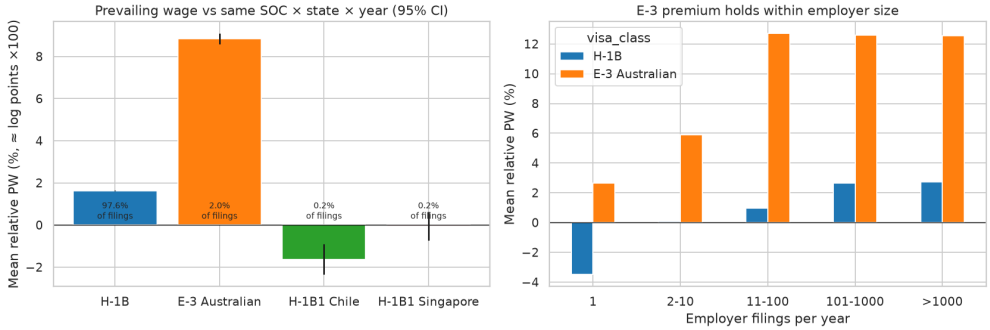


Figure 2: E-3 filings sit $+8.8\%$ above the median for the same job, state and year, H-1B $+1.6\%$, and the gap holds at every employer size.

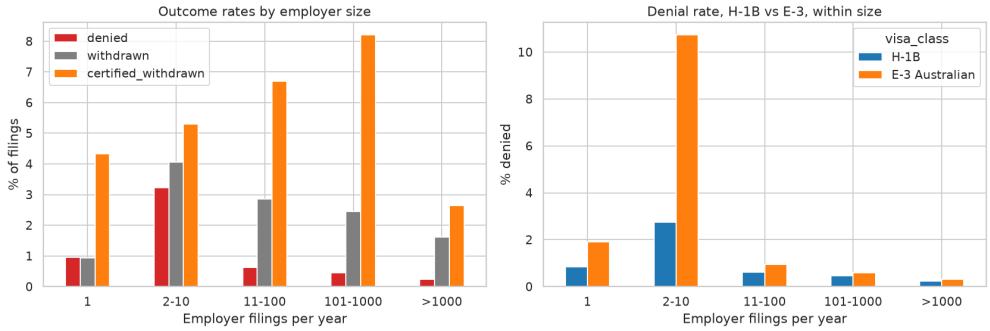


Figure 3: Denials are highest at employers with 2–10 filings a year, yet E-3 is denied more in every size group.

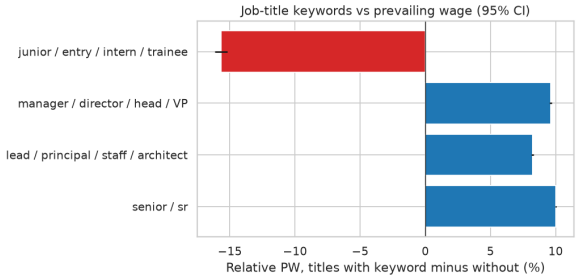


Figure 4: "Senior" titles sit $+10\%$ above others in the same job and "junior" -16%, so the relative wage shows the pay level an employer chose.

Power, checked before the run (the chance of finding a gap the size of the MID): T1 1.00, T2 0.999, T3 0.16, T4 0.93. It uses each test's SE on the exploration set, scaled up $\times$1.99 because the test set has 4$\times$ fewer employers, at $\alpha = 0.0125$ (the strictest level Holm can apply).

Test 1 (P1): do E-3 filings have a different prevailing wage for the same job, state and year?

a) Hypotheses, written before running the test

H_0	$\beta = 0$
H_1	$\beta \neq 0$
Significance level	$\alpha = 0.05$ overall (Holm, 4 tests)
Side	two-sided
Test	t-test on β, one level per occupation $\times$ state $\times$ year, SE grouped by employer
Smallest difference that matters	5% ($\lvert\beta\rvert \geq \log 1.05 = 0.049$)

y = log yearly prevailing wage, so β is the E-3 minus H-1B gap in log points (≈%). Two-sided, because P1 and P2 ask whether the groups differ in either direction. 5%, because it is about $4,400 at the median wage, about one year of pay growth here, and half the gap between "senior" and other titles in the same job: a smaller gap wouldn't put a worker at a different pay level.

b) Other tests we considered

Two groups and a symmetric number outcome suggest a two-sample t-test, but the groups must be compared within the same job, state and year (Fig. 2), and filings aren't independent. A regression with one level per cell makes all 2,729 within-cell comparisons at once, and grouping the SE by employer handles the near-copies.

Also considered	Why we did not use it
Welch t-test	compares the relative wage of the two groups, but treats filings as independent, so its SE (0.0027) is 2.4× too small. Run as a check: $t = 20.7$, same decision
Mann–Whitney U test	compares ranks, which can't be matched to a 5% MID; log wage is already symmetric, so ranks add nothing; also treats filings as independent
Two-group comparison	comparing all E-3 with all H-1B filings mixes differences in job, state and year into the gap

c) Result

Rows used	408,671 (10,433 E-3), 19,299 employers
Estimate	$+9.1\%$ ($\hat{\beta} = 0.087$; raw gap $+9.6\%$)
95% confidence interval	$+7.7\%$ to $+10.5\%$
Test statistic (d.f.)	$t = 13.66$ (19,298)
p-value (raw, Holm-adjusted)	3.0×10^{-42} (8.9×10^{-42})
Decision	reject H_0
Verdict	meaningful: the whole CI is above 5%

For the same job, state and year, E-3 filings have a prevailing wage about 9% higher than H-1B filings, well past the 5% that matters. Job, state and year explain almost none of the raw $+9.6\%$ gap. The result is steady across years ($+8$ to $+12\%$; not planned in advance, see `report/inference_results.md`).

Test 2 (P1): does the wage gap remain within the same employer?

a) Hypotheses, written before running the test

H_0	$\beta = 0$
H_1	$\beta \neq 0$
Significance level	$\alpha = 0.05$ overall (Holm)
Side	two-sided
Test	as Test 1, cells = employer $\times$ occupation $\times$ state $\times$ year
Smallest difference that matters	5%

Only employers that file both visas for the same job, state and year count. If the Test 1 gap disappeared here, it would only mean that higher-paying employers hire Australians; if it remains, the same employer files the two groups differently.

b) Other tests we considered

Holding the employer fixed removes its pay policy, industry and lawyer, which Test 1 leaves open; one level per cell does this without assuming anything about how employers differ.

Also considered	Why we did not use it
Paired t-test	counts each employer once instead of each filing. Run as a check with one gap per employer (695): $+5.3\%$ (95% CI $+3.7\%$ to $+7.0\%$, includes 5%), $t = 6.59$, $p = 9 \times 10^{-11}$
Wilcoxon signed-rank test	tests the median employer, not the average gap in %, so it can't be compared with the 5% MID. Run as a check: median gap only $+1.5\%$ ($p = 3 \times 10^{-12}$); 55% of employers file E-3 higher
Matching	pairing each E-3 filing with an H-1B filing throws away most H-1B filings without removing more bias

c) Result

Rows used	71,060 (3,356 E-3), 695 employers, 1,906 cells
Estimate	$+6.8\%$ ($\hat{\beta} = 0.065$)
95% confidence interval	$+5.3\%$ to $+8.3\%$
Test statistic (d.f.)	$t = 9.24$ (694)
p-value (raw, Holm-adjusted)	2.9×10^{-19} (5.8×10^{-19})
Decision	reject H_0
Verdict	meaningful: the whole CI is above 5%

The same employer files its E-3 workers at a prevailing wage about 7% higher than its H-1B workers in the same job, state and year: about 75% of the Test 1 gap (from a different set of employers, those filing both visas). But the gap is **concentrated in some employers**: counting each employer once gives $+5.3\%$ and the median employer only $+1.5\%$, and without copy rows it is $+6.0\%$; all three CIs include 5%. So the same employer does file E-3 higher, but only the filing-weighted average clearly passes 5%; for a typical employer the gap is small.



Figure 5: All four results with 95% CIs; the grey band is ±MID. Tests 1–3 clear it; Test 4's CI overlaps it.

Test 3 (P2): are E-3 filings denied more often by employers of the same size?

a) Hypotheses, written before running the test

H_0	$\beta = 0$
H_1	$\beta \neq 0$
Significance level	$\alpha = 0.05$ overall (Holm)
Side	two-sided
Test	regression on denied (1/0), one level per employer-size group × year, SE grouped by employer
Smallest difference that matters	0.5 percentage points

y = 1 if denied, 0 if not. Size groups: 1, 2–10, 11–100, 101–1000, >1000 filings a year; all six years are used. 0.5 pp, because the H-1B denial rate is 0.78% (exploration set), so it means at least 1.6× the risk, and it is over half of the whole FY2017–22 fall in denials (1.36% → 0.49%).

b) Other tests we considered

Employer size affects denials the most (Fig. 3), so the comparison is within size group and year. A regression on a 1/0 outcome (a *linear probability model*) gives the gap directly in percentage points, the unit of the MID, with no extra assumption about its shape.

Also considered	Why we did not use it
CMH test	the Cochran–Mantel–Haenszel test, the classic test for 2×2 tables within groups; it gives an odds ratio, not pp, and treats filings as independent. Run as a check: odds ratio 3.82, $p < 10^{-100}$
Logistic regression	gives an odds ratio, not pp. Run as a check, same cells and grouped SE: odds ratio 3.80, $z = 24.9$
χ^2 test	on the raw 2×2 table it ignores employer size, which is what the question is about

c) Result

Rows used	791,227 (14,587 E-3), 32,956 employers
Estimate	$+3.79$ pp (E-3 5.32%, H-1B 0.77% raw; raw gap $+4.55$ pp)
95% confidence interval	$+3.26$ to $+4.31$ pp
Test statistic (d.f.)	$t = 14.12$ (32,955)
p-value (raw, Holm-adjusted)	4.0×10^{-45} (1.6×10^{-44})
Decision	reject H_0
Verdict	meaningful: the whole CI is above 0.5 pp

Among employers of the same size in the same year, E-3 filings are denied 3.8 pp more often. Employer size explains only 17% of the raw gap: it shrinks the gap but doesn't reverse it, so there is no Simpson's paradox. Planned power was low (0.16) because the expected SE (0.33 pp) was large next to a 0.5 pp gap at $\alpha = 0.0125$. The real SE was 0.27 pp and the gap 3.79 pp, about eight times the MID, so low power didn't limit this result. *Not planned in advance:* the gap shrinks from $+6.6$ pp in FY2018 to $+1.6$ pp in FY2022.

Test 4 (P2): does the denial gap remain within the same employer?

a) Hypotheses, written before running the test

H_0	$\beta = 0$
H_1	$\beta \neq 0$
Significance level	$\alpha = 0.05$ overall (Holm)
Side	two-sided
Test	as Test 3, cells = employer × year
Smallest difference that matters	0.5 percentage points

Only employers filing both visas in the same year count. Holding the employer fixed removes its experience with the process and how it fills in forms, which size groups only roughly capture.

b) Other tests we considered

Also considered	Why we did not use it
CMH test	by employer × year, the standard test for a 2×2 gap within many groups, but gives an odds ratio, not pp. Run as a check: odds ratio 1.89, $p = 7.6 \times 10^{-6}$
Paired t-test	one gap per employer (1,547 employers), no regression. Run as a check: $+0.59$ pp (95% CI $+0.10$ to $+1.08$), $p = 0.018$
Wilcoxon signed-rank test	tests the median employer, and only 306 of 1,547 employers have any gap, so the median is 0. Run as a check: $p = 0.0005$
Logistic regression	with one column per employer (1,547 columns), but most employer-years have no denial at all, so the odds ratios come out biased

c) Result

Rows used	382,911 (8,003 E-3), 1,547 employers
Estimate	$+0.51$ pp (H-1B rate at these employers 0.30%)
95% confidence interval	$+0.22$ to $+0.81$ pp
Test statistic (d.f.)	$t = 3.44$ (1,546)
p-value (raw, Holm-adjusted)	5.9×10^{-4} (5.9×10^{-4})
Decision	reject H_0
Verdict	possibly too small: the CI includes 0.5 pp

Within the same employer the gap is real but small: we can't tell whether it reaches the 0.5 pp that matters. Compared with these employers' low H-1B rate (0.30%), E-3's rate is about 2.7× as high. Holding the employer fixed removes 86% of the Test 3 gap (different sets of employers, as in Test 2), so most of the E-3 denial gap comes from *which* employers file E-3, not from how one employer's filings are treated.

Summary: main tests and the other tests we ran

Every other test agrees on the direction and the decision (reject H_0). The "vs MID" column says whether its 95% CI clears the MID (yes), includes it (overlaps) or has no CI.

Test	Test used	Result	vs MID
T1	main regression	$+9.1\%$ (7.7 to 10.5)	yes
	same, without copy rows	$+8.4\%$ (7.1 to 9.6)	yes
	Welch t-test	$+5.7\%$, $p = 2 \times 10^{-93}$	no CI
T2	main regression	$+6.8\%$ (5.3 to 8.3)	yes
	same, without copy rows	$+6.0\%$ (4.7 to 7.4)	overlaps
	paired t-test	$+5.3\%$ (3.7 to 7.0)	overlaps
	Wilcoxon signed-rank test	median $+1.5\%$, $p = 3 \times 10^{-12}$	no CI
T3	main regression	$+3.79$ pp (3.26 to 4.31)	yes
	same, without copy rows	$+3.67$ pp (3.17 to 4.18)	yes
	CMH test	odds ratio 3.82, $p < 10^{-100}$	no CI
	logistic regression	odds ratio 3.80, $p = 3 \times 10^{-137}$	no CI
T4	main regression	$+0.51$ pp (0.22 to 0.81)	overlaps
	same, without copy rows	$+0.48$ pp (0.16 to 0.79)	overlaps
	CMH test	odds ratio 1.89, $p = 8 \times 10^{-6}$	no CI
	paired t-test	$+0.59$ pp (0.10 to 1.08)	overlaps
	Wilcoxon signed-rank test	median 0, $p = 0.0005$	no CI

Tests 1 and 3 hold under every check. Test 4 stays "possibly too small". Test 2 is the fragile one: only the main, filing-weighted test clears 5%, so its premium is concentrated in some employers.

6. Limitations

- **Association, not cause** (Section 4): we show that filings differ, not that the visa causes it.
- **Prevailing wage, not pay received.** It is the legal minimum, and the experience-level column was removed from this version of the data; the relative wage shows it only indirectly (Fig. 4). State is coarser than the metro area DOL actually uses.
- **FY2019 left out of the wage tests**, since its file has no prevailing wage, and most of its work locations are blank.
- **Copies kept.** If some identical rows are export errors, big employers count too much. Re-running without them keeps every decision, but Test 2's CI then includes 5% (summary table).
- **Tests 2 and 4 cover only employers filing both visas**, which are larger than average.
- **Denial here is DOL's check of the form**, not the visa decision, which USCIS makes later.
- **Extras not planned in advance.** The year-by-year results and the employer-level checks of Tests 2 and 4 and the runs without copy rows were not in the plan and are not Holm-corrected. The full test plan, and what changed in it, is in `report/preregistration.md`.